

Integration by Parts

Today's Goals

- ◆ You should be able to explain the process of integration by parts in relation to the Product Rule.
- ◆ You should be able to use integration by parts to evaluate definite and indefinite integrals.
- ◆ You should be able to determine whether integration by parts or integration by substitution is helpful in evaluating an integral.

1. Calculate $\int x \cdot \sin x^2 dx$. Can we use substitution to calculate $\int x \cdot \sin x dx$?

Solution

Recall that if we take $u = x^2$, then $du = 2x dx$ so $x dx = \frac{1}{2} du$. Therefore,

$$\int x \cdot \sin x^2 dx = \frac{1}{2} \int \sin(u) du = \frac{-\cos(u)}{2} + C = \frac{-\cos(x^2)}{2} + C.$$

On the other hand, we cannot evaluate $\int x \cdot \sin x dx$ via substitution. Using $u = x$ gets us nowhere, and $u = \sin x$ would require $du = \cos x dx$, but we have no cosine in our integral. Therefore, there are only a limited number of possibilities for u and none of them work.

2. Integration by substitution allowed us to “undo” the chain rule. In this problem you are going to learn how to “undo” the product rule.

- (a) State the product rule, in both functional and Leibniz notation:

$$\begin{aligned} \frac{d}{dx}(u(x) \cdot v(x)) &= \underline{u'(x)v(x) + v'(x)u(x)} \\ &= \underline{v \frac{du}{dx} + u \frac{dv}{dx}} \end{aligned}$$

- (b) Note that u and v are both functions of x . Now, integrate both sides of the functional notation version above with respect to x , and rearrange to find a formula for $\int u(x)v'(x)dx$. Hint: Recall the Fundamental Theorem of Calculus II!

Similarly, take the integral with respect to x of both sides of the Leibniz version and rearrange to find $\int u \frac{dv}{dx} dx$.

Solution

We first demonstrate this with functional notation:

Integrating both sides of $\frac{d}{dx}(u(x) \cdot v(x)) = u'(x)v(x) + v'(x)u(x)$ with respect to x gives us:

$$\int \frac{d}{dx}(u(x) \cdot v(x)) dx = \int v(x)u'(x) dx + \int u(x)v'(x) dx$$

and we note that $\int \frac{d}{dx}(u(x) \cdot v(x))dx = u(x) \cdot v(x)$ by FTOC I. Rearranging gives us

$$\int u(x)v'(x)dx = u(x)v(x) - \int v(x)u'(x)dx.$$

And, if we use the notation that the differential $du = u'(x)dx$, this gives us the shorthand expression: $\int u dv = uv - \int v du$

We can also demonstrate this formula using Leibniz notation: Integrating both sides with respect to x gives us:

$$\int \frac{d}{dx}(u \cdot v)dx = \int \left(v \frac{du}{dx} + u \frac{dv}{dx} \right) dx$$

The integral of the sum is the sum of the integrals:

$$\int \frac{d}{dx}(u \cdot v)dx = \int v \frac{du}{dx} dx + \int u \frac{dv}{dx} dx$$

FTOC I allows us to evaluate the left hand side:

$$uv = \int v \frac{du}{dx} dx + \int u \frac{dv}{dx} dx$$

And, if we use the notation that the differential $du = u'(x)dx$, this gives

$$uv = \int v du + \int u dv$$

Integration by Parts

If $u(x), v(x)$ are both differentiable functions of x , then

$$\int u(x) \cdot v'(x) dx = \underline{u(x)v(x) - \int v(x)u'(x)dx}.$$

We often write $u'(x)dx = du$ and $v'(x)dx = dv$, in which case the formula is

$$\int u dv = \underline{uv - \int v du}.$$

3. In this problem you will calculate $\int x \cdot \sin x dx$.

- (a) If we want to calculate $\int x \cdot \sin x dx$ using the technique of integration by parts, we must interpret it as an integral of the form $\int u(x) \cdot v'(x) dx$. What should we make u and what should make $\frac{dv}{dx}$? Is this the only choice?

Solution

Notice that we can take both derivatives and anti-derivatives of both $x, \sin(x)$, so you have choices. On the other hand, if we let $u = x$, then $\frac{du}{dx} = 1$. This will make things easier for us. So we should choose $\frac{dv}{dx} = \sin(x)$. This means that $v(x) = -\cos(x)$.

- (b) Use the formula for integration by parts and your choices for u and v to calculate $\int x \cdot \sin x \, dx$.

Solution

The IBP formula is

$$\int u(x)v'(x) \, dx = uv - \int vu'(x) \, dx$$

Here, we have $u(x) = x, u'(x) = 1, v(x) = -\cos(x), v'(x) = \sin x$, so

$$\int x \sin(x) dx = -x \cos(x) - \int -\cos(x) dx$$

$$\int x \sin(x) dx = -x \cos(x) + \sin(x)$$

4. Evaluate the following integrals, using integration by parts. Clearly state what you are choosing for u and what you are choosing for v .

(a) $\int x \cdot \ln x \, dx$

Solution

This is not an integral that we simply know the antiderivative of. Also, our other technique of substitution fails here since no choice of u helps with simplifying this integral! This is where Integration by Parts (IBP) comes in useful!

$$\int u(x)v'(x) \, dx = u(x)v(x) - \int v(x)u'(x) \, dx$$

How can we see $\int \ln x \cdot x \, dx$ as an integral of the type $\int u(x)v'(x) \, dx$? Let's try matching functions - currently, we only know the derivative of $\ln x$ and not its antiderivative, so that forces $u(x) = \ln x$. Thus we have:

$$u(x) = \ln x \quad v'(x) \, dx = x \, dx$$

Then

$$u'(x) = \frac{1}{x} \quad v(x) = \frac{x^2}{2}$$

Putting it all together:

$$\begin{aligned} \int \ln x \cdot x \, dx &= \ln x \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \cdot \frac{1}{x} \, dx \\ &= \ln x \cdot \frac{x^2}{2} - \int \frac{x}{2} \, dx \\ &= \ln x \cdot \frac{x^2}{2} - \frac{x^2}{4} + C \end{aligned}$$

Let's check our answer by differentiating:

$$\begin{aligned}\frac{d}{dx} \left(\ln x \cdot \frac{x^2}{2} - \frac{x^2}{4} + C \right) &= \frac{1}{x} \cdot \frac{x^2}{2} + \ln x \cdot x - \frac{x}{2} + 0 \\ &= \frac{x}{2} + \ln x \cdot x - \frac{x}{2} \\ &= \ln x \cdot x\end{aligned}$$

That works!

(b) $\int x e^x dx$

Solution

Let's sub $u = x$, since it simplifies on differentiation, and sub $dv = e^x dx$ since we know the anti-derivative of $e^x = e^x$. Then $du = 1dx$ and $v = e^x$:

$$\begin{aligned}\int x e^x dx &= x e^x - \int e^x dx \\ &= \boxed{x e^x - e^x + C}\end{aligned}$$

(c) $\int_0^1 x e^x dx$

Solution

To compute definite integrals that would involve IBP, we start by first computing the indefinite integral, and then plugging in the bounds. In this case, from part (b), we already know that

$$\int x e^x dx = x e^x - e^x + C$$

So:

$$\begin{aligned}\int_0^1 x e^x dx &= (x e^x - e^x)|_0^1 \\ &= (e - e) - (0 - e^0) \\ &= \boxed{1}\end{aligned}$$

(d) $\int x^2 e^x dx$

Solution

By the same reasoning as in part (b), let $u = x^2$ and $\frac{dv}{dx} = e^x$. Then $\frac{du}{dx} = 2x$ and $v = e^x$. Then our integration by parts formula gives

$$\int x^2 e^x dx = x^2 e^x - \int 2x e^x dx$$

Notice we need to do integration by parts again to calculate $\int 2x e^x dx$. Fortunately we have

already done this in the previous problem.

$$\int x^2 e^x dx = x^2 e^x - 2(xe^x - e^x) + c$$

(e) $\int \ln x dx$

Solution

This doesn't look like an IBP problem. However, we can't use substitution here, and so let's give IBP a shot anyway! Notice that we can rewrite

$\int \ln x dx = \int 1 \cdot \ln x dx$ So, let $u = \ln x$, since it simplifies on differentiation, and $v'(x) dx = 1 dx$, or $dv = dx$. Then $du = \frac{1}{x} dx$ and $v = x$: So,

$$\begin{aligned} \int \ln x dx &= \ln x \cdot x - \int x \frac{1}{x} dx \\ &= x \ln x - \int dx \\ &= \boxed{x \ln x - x + C} \end{aligned}$$

(f) $\int_0^{\ln 3} x^2 e^{2x} dx$

Solution

Let's focus on computing $\int x^2 e^{2x} dx$ first. Let $u = x^2$, since it simplifies on differentiation, and $dv = e^{2x} dx$ since we know the anti-derivative, and it isn't much more difficult than dv . Then $du = 2x dx$ and $v = \frac{e^{2x}}{2}$: So,

$$\begin{aligned} \int x^2 e^{2x} dx &= x^2 \cdot \frac{e^{2x}}{2} - \int \frac{e^{2x}}{2} 2x dx \\ &= \frac{x^2 e^{2x}}{2} - \int x e^{2x} dx \end{aligned}$$

But to compute $\int x e^{2x} dx$ we need IBP again! Deep breaths. We got this. Let $u = x$, since it simplifies on differentiation, and $dv = e^{2x} dx$ since we know the anti-derivative, and it isn't much more difficult than dv . Then $du = 1 dx$ and $v = \frac{e^{2x}}{2}$:

$$\begin{aligned} \int x e^{2x} dx &= x \cdot \frac{e^{2x}}{2} - \int \frac{e^{2x}}{2} dx \\ &= \frac{x e^{2x}}{2} - \int e^{2x} dx \\ &= \frac{x e^{2x}}{2} - \frac{e^{2x}}{2} + C \end{aligned}$$

Finally, we have that

$$\int x^2 e^{2x} dx = \frac{x^2 e^{2x}}{2} - \left(\frac{x e^{2x}}{2} - \frac{e^{2x}}{2} + C \right) = \frac{x^2 e^{2x}}{2} - \frac{x e^{2x}}{2} + \frac{e^{2x}}{2} - C$$

So:

$$\begin{aligned} \int_0^{\ln 3} x^2 e^{2x} dx &= \left(\frac{x^2 e^{2x}}{2} - \frac{x e^{2x}}{2} + \frac{e^{2x}}{2} \right) \Big|_0^{\ln 3} \\ &= \left(\frac{(\ln 3)^2 e^{2 \ln 3}}{2} - \frac{\ln 3 e^{2 \ln 3}}{2} + \frac{e^{2 \ln 3}}{2} \right) - \frac{e^0}{2} \\ &= \boxed{\frac{(\ln 3)^2 e^{2 \ln 3}}{2} - \frac{\ln 3 e^{2 \ln 3}}{2} + \frac{e^{2 \ln 3}}{2} - \frac{1}{2}} \end{aligned}$$

5. Evaluate the following integrals. Note: different techniques apply!

(a) $\int \frac{1}{0.5x - 3} dx.$

Solution

$2 \ln |x - 6| + C.$ Substitute $u = 0.5x - 3$, or rewrite the integrand as $\frac{1}{0.5} \cdot \frac{1}{x-6}.$

(b) $\int_0^{\sqrt{15}} 5x(x^2 + 1)^{-9/4} dx.$

Solution

$\frac{31}{16}.$ Substitute $u = x^2 + 1.$ You'll need to simplify $16^{-5/4}$; the easiest way is to rewrite it as $\frac{1}{16^{5/4}} = \frac{1}{(16^{1/4})^5}.$

(c) $\int \left(\sqrt[3]{x} + \frac{1}{\sqrt[3]{x^2}} \right)^3 dx.$

Solution

$$\int x + 3 + 3x^{-1} + x^{-2} dx = \frac{x^2}{2} + 3x + 3 \ln |x| - x^{-1} + C.$$

(d) $\int \frac{4x + 4}{2x + 1} dx.$

Solution

$2x + \ln |2x + 1| + C.$ Substitute $u = 2x + 1.$

(e) $\int \frac{x + 2}{x^2 + 4x - 5} dx.$

Solution

$\frac{1}{2} \ln |x^2 + 4x - 5| + C$. Substitute $u = x^2 + 4x - 5$.

(f) $\int_{-10}^{10} \sin(x^3) dx$.

Solution

Since $\sin(x^3)$ is an odd function and the interval is symmetric around 0, the integral is zero. (Note that the antiderivative of $\sin(x^3)$ is not an elementary function.)

(g) $\int x \arctan x dx$.

Solution

$-\frac{x}{2} + \frac{\arctan x}{2} + \frac{1}{2}x^2 \arctan x + C$. Integrate by parts with $u = \arctan x$ and $dv = x dx$.

(h) $\int x^2 \cos x dx$.

Solution

$x^2 \sin x + 2x \cos x - 2 \sin x + C$. Integrate by parts twice.